

Model Order Reduction and Hyperreduction for PDEs

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The following are notes on model order reduction and hyperreduction for PDEs.

1 Empirical Interpolation Method (EIM)

1.1 Notation

- $(X, \langle \cdot, \cdot \rangle_X)$ is a Hilbert space.
- $\mathcal{D}$: Parameter space. We assume that $\mathcal{D} \subset \mathbb{R}^{p_d}$ is a compact set.
- $\mathcal{N}$: Dimension of the High-dimensional model (HDM) or High-fidelity model.
- $(\phi_i)_{i=1}^{\mathcal{N}}$: Basis of the high-fidelity approximation space $X^{\mathcal{N}}$.
- N : Dimension of the reduced model.
- $(\xi_i)_{i=1}^N$: Basis of the reduced approximation space $X^N \subset X^{\mathcal{N}}$. Typically, $\xi_i := u^{\mathcal{N}}(\mu_i)$ is a snapshot, where μ_i is a selected parameter.

1.2 The method

1.2.1 Overview

The empirical interpolation method (EIM) is a technique that aims to interpolate a function $g : \mathbb{R}^d \times \mathcal{D} \rightarrow \mathbb{R}$ where each $g(\cdot, \mu)$ lives in a Banach space (usually an L^p space, with $p \in [1, \infty]$). The goal is to find an approximation of g that is affine in the parameter μ , ie. of the form,

$$g(\cdot, \mu) \approx g_M(\cdot, \mu) = \sum_{m=1}^M \beta_m(\mu) q_m(\cdot), \quad (1)$$

where q_m are basis functions and β_m are the coefficients that depend on the parameter μ .

1.2.2 Example application

As discussed below in 1.3.1, the EIM is particularly useful in the context of reduced-order modeling (ROM) for PDEs, where the separability assumption (S) is important to efficiently compute the reduced-order model (ROM) during the online phase. In the many cases where (S) is not satisfied, the EIM can be used to approximate the operator a_μ by an affine function of the form (1).

For example, in the case of a second-order elliptic coercive problem with a bilinear form a_μ of the form,

$$a_\mu(u, v) = \int_{\Omega} \nabla u \cdot \nabla v + \int_{\Omega} g_\mu uv, \quad (2)$$

where $g_\mu : \Omega \rightarrow \mathbb{R}$ is a parameter-dependent function, the EIM can be used to approximate the function g_μ by a sum of separable functions as in (1), such that it is possible to approximate the bilinear form a_μ as,

$$a_\mu(u, v) \approx a_\mu^M(u, v) = \int_{\Omega} \nabla u \cdot \nabla v + \sum_{m=1}^M \beta_m(\mu) \int_{\Omega} q_m uv, \quad (3)$$

which satisfies the separability assumption (S) and allows for efficient computation of the reduced-order model (ROM) during the online phase.

1.2.3 The EIM algorithm

The goal is to construct an approximation g_M of the function g ,

$$g_M(x, \mu) = \sum_{m=1}^M \beta_m(\mu) q_m(x). \quad (4)$$

We will see that although we will explicitly construct the functions $x \mapsto q_m(x)$, the functions $\mu \mapsto \beta_m(\mu)$ will be implicitly defined as the solution of a well-posed linear system of dimension M that depends on the particular parameter μ we want to evaluate the approximation at.

In practice, the approximation g_M is obtained through a two-step process, of which we give a brief intuition next.

Method intuition The first step is to select a set of parameters μ_i from the parameter space $\mathcal{D}$. These are selected through a greedy procedure, where we select the parameters μ_i that maximize the error of the approximation. We thus obtain a set of M snapshots $(x \mapsto g_{\mu_i}(x))_{i=1}^M$. At this point, a first naïve idea would be to simply look for a g^M of the form,

$$g^M(x, \mu) = \sum_{i=1}^M \beta_i(\mu) g_{\mu_i}(x), \quad (5)$$

where the g_{μ_i} are simply the q_i , and the β_i are some coefficients that depend on the parameter μ .

Now say we want to evaluate the approximation g^M at a particular parameter point μ . We would need to determine the coefficients $\beta_i(\mu)$. In the scope of this method, this is done by closing the system (5) by imposing M conditions for M different points x_i in the domain Ω , and solving the resulting linear system. These points $(x_i)_{i=1}^M$ are also selected through a greedy procedure.

To close the system, we choose to impose that the approximation (5) must be exact at the points $(x_i)_{i=1}^M$, ie.

$$g^M(x_i, \mu) = g(x_i, \mu), \quad \forall i \in \{1, \dots, M\}. \quad (6)$$

However, depending on the choice of the points $(x_i)_{i=1}^M$, the linear system could *a priori* be ill-defined (ie. redundant). This is what will motivate the transformation of the basis functions g_{μ_i} into the basis functions q_i that are built from the selected points $(x_i)_{i=1}^M$ in a way that makes the linear system clearly well-defined.

Details of the method Let's now go into the details of the method. For a different perspective on the method, see [1].

1. **Greedy selection of the parameters μ_i** : We select a set of M parameters $(\mu_i)_{i=1}^M$ from the parameter space $\mathcal{D}$, such that each new g_{μ_i} is as different as possible from the previous ones. This is simply done by solving the optimization problem, starting with an arbitrarily chosen μ_1 ,

$$\mu_i = \operatorname{argsup}_{\mu \in \mathcal{D}} \inf_{w \in W_{i-1}} \|g(\cdot, \mu) - w\|_X, \quad \forall i \in \{2, \dots, M\}, \quad (7)$$

where $W_{i-1} = \operatorname{span}(g_{\mu_1}, \dots, g_{\mu_{i-1}})$ is the space spanned by the previous g_{μ_j} , and $\|\cdot\|_X$ is the norm in the Hilbert space X (which is typically an L^p space). We thus obtain the set of $\{g_{\mu_i}\}_{i=1}^M$.

2. **Greedy selection of the points x_i and construction of the basis functions q_i** : We now need to select a set of M points $(x_i)_{i=1}^M$ in the domain Ω such that (6) is satisfied. We start by setting $x_1 = \operatorname{argsup}_{x \in \Omega} |g_{\mu_1}(x)|$ (assume that g_{μ} is $\mathcal{C}^0$ or L^∞ for all μ), and define $q_1(x) = \frac{g_{\mu_1}(x)}{g_{\mu_1}(x_1)}$. Now, for $i \in \{2, \dots, M\}$, we start by finding the coefficients $\sigma \in \mathbb{R}^i$ that satisfy the linear system,

$$\sum_{j=1}^{i-1} \sigma_j q_j(x_k) = g_{\mu_i}(x_k), \quad \forall k \in \{2, \dots, M\}, \quad (8)$$

The point x_i is then selected as,

$$x_i = \operatorname{arg\,esssup}_{x \in \Omega} \left| g_{\mu_i}(x) - \sum_{j=1}^{i-1} \sigma_j q_j(x) \right|, \quad \forall i \in \{2, \dots, M\}, \quad (9)$$

which is the spatial point, at iteration i , where the approximation error of g_{μ_i} with respect to itself is maximal.

Then, we define the basis functions q_i as,

$$q_i(x) = \frac{g_{\mu_i}(x) - \sum_{j=1}^{i-1} \sigma_j q_j(x)}{g_{\mu_i}(x_i) - \sum_{j=1}^{i-1} \sigma_j q_j(x_i)}, \quad \forall i \in \{2, \dots, M\}, \quad (10)$$

where we can see that the $q_i(x_i) = 1$ and $q_i(x_j) = 0$ for $j < i$ as ensured by (8).

1.2.4 Some properties of the EIM

- The span of the basis functions $(q_i)_{i=1}^M$ is the same as the span of the functions $(g_{\mu_i})_{i=1}^M$, ie.

$$\mathbb{V}_M = \operatorname{span}(q_1, \dots, q_M) = \operatorname{span}(g_{\mu_1}, \dots, g_{\mu_M}).$$

- When evaluating the approximation (4) at a particular parameter μ , we need to solve the linear system,

$$\sum_{m=1}^M \beta_m(\mu) q_m(x_k) = g(x_k, \mu), \quad \forall k \in \{1, \dots, M\}, \quad (11)$$

which has a unique solution, since the matrix $[B]_{ij} = q_i(x_j)$ is lower triangular with diagonal entries $q_i(x_i) = 1$.

- The invertibility of the matrix $[B]_{ij} = q_i(x_j)$ also implies that for any M -dimensional vector $\alpha \in \mathbb{R}^M$, there exists a unique $w \in \mathbb{V}_M$ such that,

$$w(x_j) = \alpha_j, \quad \forall j \in \{1, \dots, M\}.$$

- The following two properties are, by construction, satisfied by the EIM:

1. The approximation (4) is exact at the points $(x_i)_{i=1}^M$, for all $\mu \in \mathcal{D}$, ie.

$$g_M(x_i, \mu) = g(x_i, \mu), \quad \forall i \in \{1, \dots, M\}, \forall \mu \in \mathcal{D}. \quad (12)$$

2. The approximation (4) is exact at the parameter points $(\mu_i)_{i=1}^M$, for all $x \in \Omega$, ie.

$$g_M(x, \mu_i) = g(x, \mu_i), \quad \forall i \in \{1, \dots, M\}, \forall x \in \Omega. \quad (13)$$

1.2.5 Error analysis

When the considered norm is the L^∞ norm, the *a priori* error analysis of the EIM involves the Lebesgue constant $\Lambda_M = \sup_{x \in \Omega} \sum_{i=1}^M |L_i(x)|$, where $L_i \in \mathbb{V}_M$ is the Lagrange polynomial associated with the q_i , ie. $L_i(x_j) = \delta_{ij}$, and δ_{ij} is the Kronecker delta. We have the bound,

$$\|g_\mu - g_\mu^M\|_{L^\infty(\Omega)} \leq (1 + \Lambda_M) \inf_{w \in \mathbb{V}_M} \|g_\mu - w\|_{L^\infty(\Omega)}. \quad (14)$$

While an upper bound for Λ_M is given by $\Lambda_M \leq 2^M - 1$, it is actually heuristically observed that Λ_M grows much slower than 2^M in practice.

1.3 Bibliography recaps

The purpose of this section is to recap the main ideas of the papers read so far.

1.3.1 An empirical interpolation method: Application to efficient reduced-basis discretization of partial differential equations (2004)

- Seminal paper introducing the empirical interpolation method (EIM).
- Places itself in the nice context of second-order elliptic coercive problems and deals with the variational formulation of the problem.
- In the case where the operator a is affine in the parameter μ , ie.

$$a_\mu(w, v) = \sum_{k=1}^K \Theta^k(\mu) a^k(w, v), \quad (S)$$

the RB method becomes very efficient, since we can assemble and compute once and for all the matrices $[A^k]_{ij} = a^k(\xi_i, \xi_j)$ once and for all, and then compute the reduced-order model (ROM) by solving the problem. Indeed, as the next paragraph will show, computing the matrices $[A^k]_{ij}$ scales with $\mathcal{N}$, and we therefore want to make all the expensive computations (that involve $\mathcal{N}$) offline, and then compute the reduced-order model (ROM) online, which only involves the reduced dimension N .

Indeed, if that (S) was not satisfied, when evaluating, during the online phase, a the reduced-order model (ROM), we would have to solve the problem,

$$\text{Find } u_\mu^N \in X^N \text{ such that } a_\mu(u_\mu^N, v^N) = l_\mu(v^N), \quad \forall v^N \in X^N, \quad (15)$$

with typically $N \ll \mathcal{N}$.

However, an expensive computation involving $\mathcal{N}$ is hidden inside!

Indeed, problem (19) is solved by inverting the matrix A_μ^N defined as $[A_\mu^N]_{ij} = a_\mu(\xi_i, \xi_j)$. Nevertheless, the coefficients $a_\mu(\xi_i^N, \xi_j^N)$ involve the basis elements ξ_i^N which live in the high-dimensional space X^N and that thus require, for accurately computing the constitutive integrals of the operator a , a number of operations scaling with $\mathcal{N}$.

Remark: One could argue that it seems unnecessary to develop $a_\mu(\xi_i, \xi_j)$ in the basis $(\phi_i)_{i=1}^{\mathcal{N}}$, ie.

$$a_\mu(\xi_i, \xi_j) = \sum_{k=1}^{\mathcal{N}} \sum_{l=1}^{\mathcal{N}} \hat{u}_\mu^k \hat{u}_\mu^l a_\mu(\phi_k, \phi_l),$$

since in both cases we have to compute, using a quadrature rule, the integrals inside the operator a_μ . So instead of computing the integrals $a_\mu(\phi_k, \phi_l)$, why not compute the integrals $a_\mu(\xi_i, \xi_j)$ directly using a quadrature rule?

However, this argument misses a crucial point: Since the ξ_i are computed on the fine mesh of dimension $\mathcal{N}$, the number of quadrature points needed to accurately compute the integrals $a_\mu(\xi_i, \xi_j)$ involves and scales with $\mathcal{N}$.

- The EIM aims at 3 things: (i) get a “good” affine approximation of the operator a_μ in the reduced space X^N , (ii) a stable and inexpensive interpolation procedure, and (iii) an effective *a posteriori* estimator of the new interpolation error.
- See 1.2 for the details of the EIM, close to what is presented in the paper.

2 Energy-based mesh sampling and weighting (ECSW)

2.1 Dimensionality reduction of nonlinear finite element dynamic models with finite rotations and energy-based mesh sampling and weighting for computational efficiency (2014)

- The article mainly deals with a numerically stable, cost-efficient dimensionality reduction of nonlinear finite element dynamic models with and without rotational DOFs.
- The article also deals with two cases : the case where the mass matrix is (i) constant or (ii) configuration-dependent.
- For this recap, we will only focus on models without rotational DOFs, and only consider Euler implicit and explicit time integration schemes. Furthermore, we will only consider the case where the mass matrix is constant.